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→ Sum Of Squared Errors

↳ When a matrix is not perfectly additive we still try to build the tree but then the tree distances and the distances in matrix are not equal.

↳ So we measure how wrong the tree is using

$$SSE = \sum_{i < j} (D[i,j] - d_t(i,j))^2$$

↓
given distance
matrix

↓
distance b/w i & j in the
constructed tree

9) $SSE = 0$, matrix is perfectly additive

$D:$	i	j	k	l
i	0	3	4	3
j	3	0	4	5
k	4	4	0	2
l	3	5	2	0

$d_t:$	i	j	k	l
i	0	3	4	(4)
j	3	0	4	(4)
k	4	4	0	2
l	4	4	2	0

$$SSE = 1^2 + 1^2 = 2$$

(i, l) (j, l)

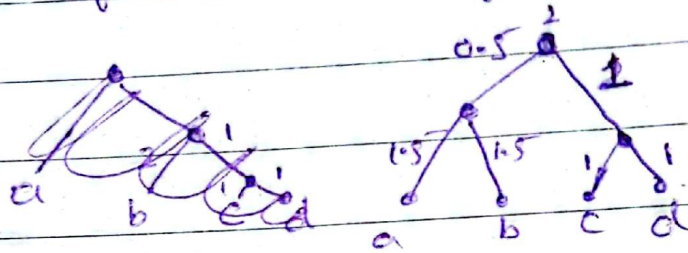
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▷ UPGMA : Clustering Heuristic

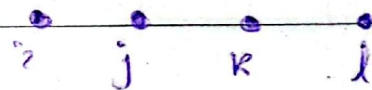
→ Ultrametric Tree:

↳ Distance from root to any leaf is same



→ Algorithm

D:	i	j	k	l
i	0	3	4	3
j	3	0	4	5
k	4	4	0	2
l	3	5	2	0

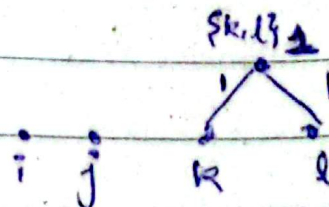


i) Select the minimum possible distance.

Here $k, l \rightarrow 2$

ii) make connection b/w two.

Distance from the root ~~should~~ to leaf should be equal.



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ii. Make cluster for that two nodes and make a new matrix.

	i	j	$\{k, l\}$
i	0	3	(3.5)
j	3	0	(4.5)
$\{k, l\}$	3.5	4.5	0

$\frac{4+3}{2} = 3.5$
 $\frac{4+5}{2} = 4.5$
 $2 \times 3 + 2 \times 3.5$

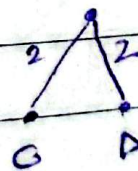
iii) Now selecting minimum again & making cluster of that. $(i, j) \rightarrow 3$

	$\{i, j\}$	$\{k, l\}$
$\{i, j\}$	0	4/2
$\{k, l\}$	4	0

$\frac{3.5+4.5}{2} = 4$

Q:

	A	B	C	D
A	0	6	8	8
B		0	8	8
C			0	(4)
D				0

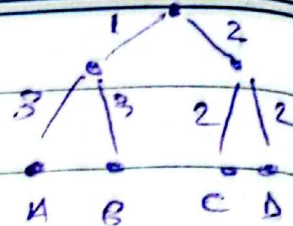


	A	B	C, D
A	0	(6)	8
B		0	8
C, D			0

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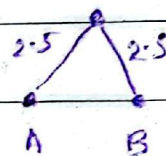
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	A, B	C, D
A, B	0	8
C, D	8	0

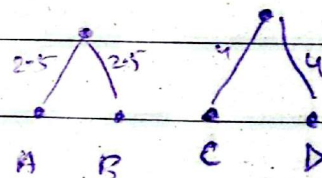


Q:

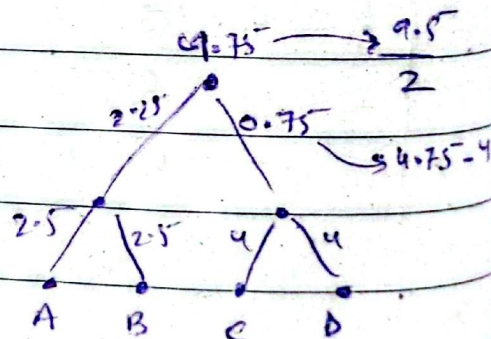
	A	B	C	D
A	0	5	9	9
B	5	0	10	10
C	9	10	0	8
D	9	10	8	0



	A, B	C	D	$\frac{9+10}{2} = 9.5$
A, B	0	9.5	9.5	
C		0	8	
D			0	



	A, B	C, D
A, B	0	9.5
C, D	9.5	0



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- Additive Phylogeny

- good - produces the tree fitting an additive matrix
- bad - fails completely on a non-additive matrix

- UPGMA

- good - produces a tree for any matrix
- bad - tree doesn't necessarily fit an additive matrix

- Neighbour Joining Algo

- good - produces the tree fitting an additive matrix
- good - provides heuristic for a non-additive matrix

▷ Neighbouring Joining Algo

① Convert additive matrix D into a neighbour joining matrix D^*

$$D^*_{i,j} = (n-2) \cdot D_{i,j} - \text{Total Distance}_i - \text{Total Distance}_j$$

D :	i	j	k	l	Total Dist _i	D^*	i	j	k	l
i	0	13	21	22	→ 56	i	0	-68	-60	-60
j	13	0	12	13	→ 38	j		0	-60	-60
k	21	12	0	13	→ 46	k			0	-68
l	22	13	13	0	→ 48	l				0

Symmetric matrix here
 rows & col sum will be equal

$$(4-2) \times 13 - 56 - 38 = -68$$

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② Neighbour-joining Theorem

↳ If D is additive, the smallest element in D^* are neighbours.

→ in D^* we have (i, j) & (k, l) are neighbours. Pick any one!

③ Calculate $\Delta_{i,j} = (\text{Total Dis}(i) - \text{Total Dis}(j)) / (n-2)$
 $= (56 - 38) / 4 - 2$
 $= 9$

④ Set $\text{Linlength}(i) = (D_{i,j} + \Delta_{i,j}) / 2$
 $\text{Linlength}(j) = (D_{i,j} - \Delta_{i,j}) / 2$

$$\text{Linb}(i) = (13 + 9) / 2 = 11$$

$$\text{Linb}(j) = (13 - 9) / 2 = 2$$

⑤ Make a new matrix D' by eliminating i^{th} & j^{th} row & column.

And add m^{th} row/col.

$$D_{k,m} = (D_{i,k} + D_{j,k} - D_{i,j}) / 2$$

⑥ Now for new matrix reapply the algo

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Q:	A	B	C	D	Total Ds
A	0	5	9	9	23
B	5	0	10	10	25
C	9	10	0	8	27
D	9	10	8	0	27

D*	A	B	C	D
A	0	-38	-32	-32
B		0	-32	-32
C			0	-38
D				0

Choose A-B

$$\Delta_{A,B} = (23 - 25) / 4 - 2 = -1$$

$$\text{Limblength (A)} = (5 - 1) / 2 = 2$$

$$\text{Limblength (B)} = (5 + 1) / 2 = 3$$

D	m	C	D	Total
m	0	(7)	(7)	14
C	7	0	8	15
D	7	8	0	15

$9 + 10 - 5 / 2 = 7$

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D^*	m	C	D
m	0	-22	-22
C		0	22
D			0

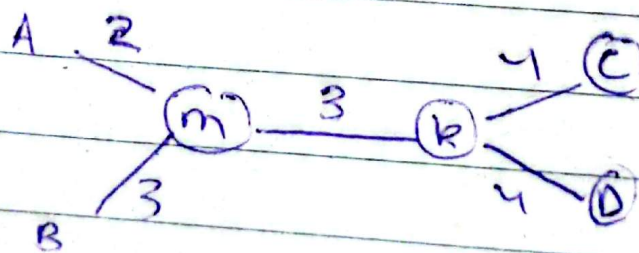
△ Choose C, D

$$\Delta_{C,D} = 0$$

$$\text{Cmbien}(C) = (8 \cdot 0) / 2 = 4$$

$$\text{Cmbien}(D) = (8 \cdot 0) / 2 = 4$$

D'	m	k	$D(m,C)$ $D(m,D)$ $D(C,D)$
m	0	(3)	$\rightarrow (7 + 7 - 8) / 2$
k	3	0	



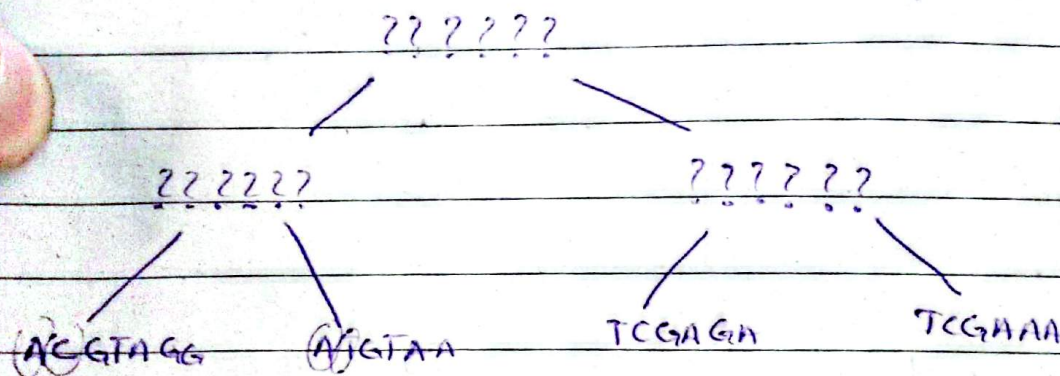
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Small Parsimony Score

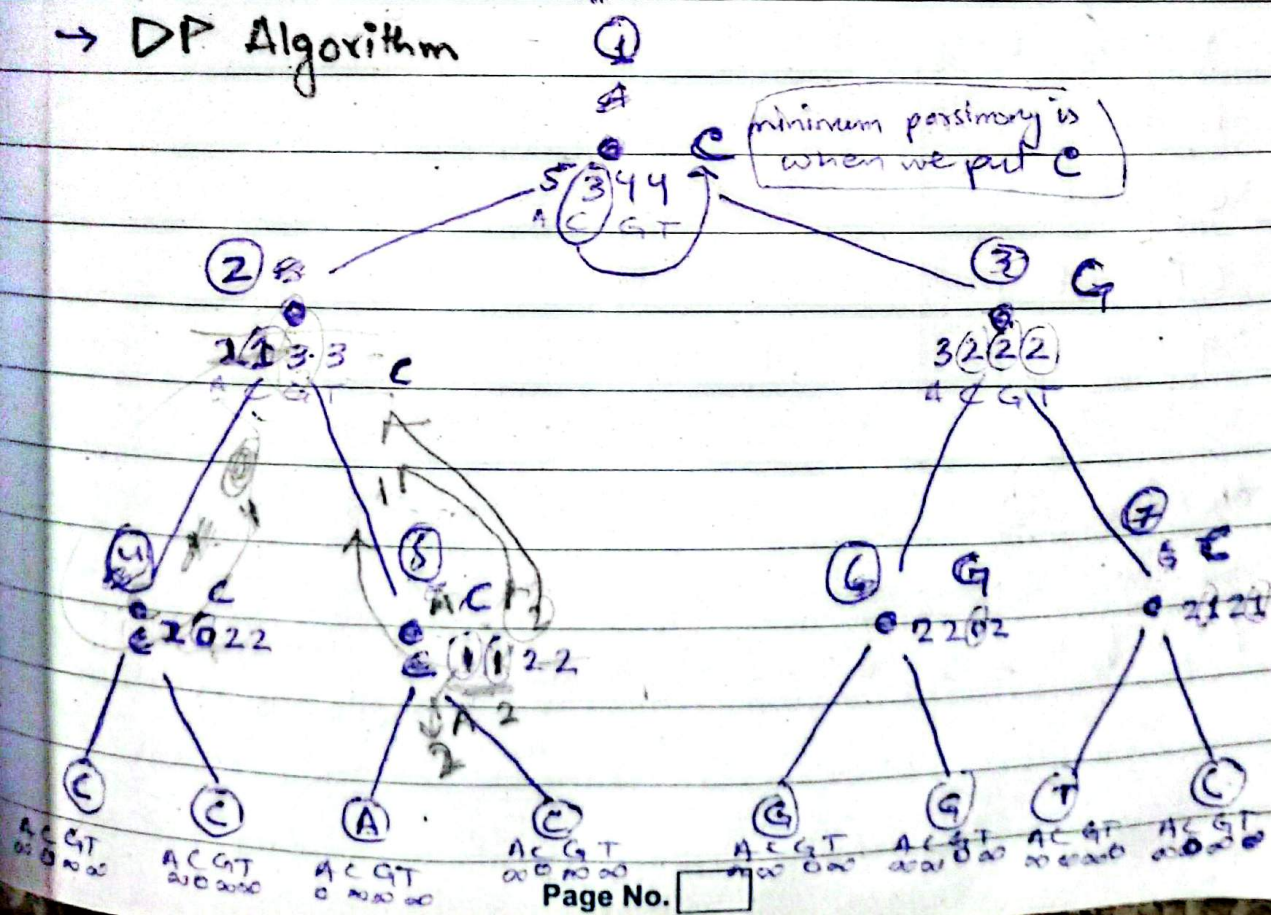
↳ Total minimum no. of character changes (mutations) in the tree

↳ It is used to guess the sequences of internal nodes in a tree



→ Parsimony Score : Sum of Hamming distance along each edge

DP Algorithm



I have Added calculations of G at node (2) &
 Date _____ A at node (1) Day _____

→ Calculation of G at Node 2

left: [2, 2, 2]

i and G equal then 0
 else 1

Check all possibilities: ↗

i	$S_i(\text{left})$	$S(i, G)$	Total	
A	2	1	3	} choose minimum
C	0	1	(1)	
G	2	0	2	
T	2	1	3	

min = 1 (coming C → G)

right: [1, 1, 2, 2]

i	$S_i(\text{right})$	$S(i, G)$	Total
A	1	1	2
C	1	1	2
G	2	0	2
T	2	1	2

min = 2

Add both sides = 1 + 1 = (2) → therefore we write 2 in G at node #02

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→ Calculation of A at Node 1

left : [2, 1, 3, 3]

A	2	0	2
C	1	1	2
G	3	1	4
T	3	1	4

min = 2

right : [3, 2, 2, 2]

A	3	0	3
C	2	1	3
G	2	1	3
T	2	1	3

min = 3

Total = 3 + 2 = 5

→ When doing backtracking to fill the nodes with characters we choose the symbol which minimizes the cost.

$$\min \{ S_i + S(i, \text{Parent}) \}$$

→

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→ For Node # 3 → parent = B C

i	Cost	$S(i, C)$	Total	
A	3	1	4	
C	2	0	(2)	✓ Chose
G	2	1	3	C at
T	2	1	3	node 3

↳ values at node 3

Complexity: $O(mnk^2)$

where,

m = no. of species

n = character in a string

K = States

▷ Large Parsimony Tree Problem

↳ In this problem unlike the small parsimony problem we are not given any tree topology.

↳ We are given only set of strings and have to calculate parsimony score for that.

↳ This is an NP-Complete Problem

↳ Because no. of trees for n string grows exponentially.

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→ Greedy Algo

- ① Start with an initial arbitrary tree
- ② Compute parsimony for that tree
- ③ For each internal edge
 - i - Generate 2 neighbour trees
 - ii - Compute parsimony score for each
- ④ If any neighbour has lower score → choose that as current tree else stop
- ⑤ Repeat until no improvement

▷ CLUSTERING

→ Markov Clustering Algo

↳ It is a graph clustering algorithm based on ~~random~~ random walks

↳ Logic : "If you randomly walk in a graph, you tend to stay longer inside dense regions (clusters) than move b/w them"

↳ Idea :

1. Expansion → Spread flow

2. Inflation → strengthen strong connections,
weaken weak ones

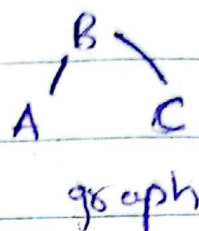
Repeat until Cluster appears.

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• Example:

1. Make an adjacency matrix



	A	B	C
A	0	1	0
B	1	0	1
C	0	1	0

2. Add 1 to diagonals

	A	B	C
A	[1	1	0]
B	[1	1	1]
C	[0	1	1]

3. Normalize columns by dividing each element by sum of its column

Column A sum = 2

Column B sum = 3

Column C sum = 2

[0.5 0.33 0]

[0.5 0.33 0.5]

[0 0.33 0.5]

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 $O(n^3)$

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4. Expansion - multiply matrix by itself (M^2) $e = 2$ (usually 2)

M^2 :
 A $\begin{bmatrix} 0.415 & 0.2739 & 0.165 \end{bmatrix}$
 B $\begin{bmatrix} 0.415 & 0.4389 & 0.415 \end{bmatrix}$
 C $\begin{bmatrix} 0.165 & 0.2739 & 0.415 \end{bmatrix}$

power
parameter

inflation parameter

5. Inflation ($\delta = 2$) $\rightarrow O(n^2)$

Square each value

	A	B	C
A	$\begin{bmatrix} 0.172 & 0.075 & 0.027 \end{bmatrix}$		
B	$\begin{bmatrix} 0.172 & 0.192 & 0.172 \end{bmatrix}$		
C	$\begin{bmatrix} 0.027 & 0.075 & 0.172 \end{bmatrix}$		

6. Normalize Columns again

A Sum $\rightarrow 0.371$ B Sum $\rightarrow 0.342$ C Sum $\rightarrow 0.371$ A $\begin{bmatrix} 0.46 & 0.22 & 0.07 \end{bmatrix}$ B $\begin{bmatrix} 0.46 & 0.56 & 0.46 \end{bmatrix}$ C $\begin{bmatrix} 0.07 & 0.22 & 0.46 \end{bmatrix}$

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↳ Here B is dominant because

↳ ~~it has higher values~~ The row of B has higher values as A & C

↳ B has strong connection to all nodes
0.46 with A & 0.46 with C

↳ A & C are weakly link with each other

↳ Random walk starts anywhere and after steps it keeps coming back to B. So B acts like a center/hub.

▷ K-CENTER CLUSTERING

↳ Farthest-First Traversal

→ It is a greedy clustering algorithm used to choose "k" cluster centers.

Points: A(0,0), B(2,2), C(8,8), D(9,9), E(18)

K = 2

1. Choose any random point : A(0,0)

2. Now compute distance of all points from A.

Point	Distance
B(2,2)	$\sqrt{(2-0)^2 + (2-0)^2} = \sqrt{8} = 2.83$
C(8,8)	$\sqrt{(8-0)^2 + (8-0)^2} = \sqrt{128} = 11.31$

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D(9,9)

$\sqrt{162} = 12.73$

E(1,8)

$\sqrt{65} = 8.06$

3. Pick the farthest point i.e. D(9,9) $\rightarrow 12.73$

4. Center #2 = D(9,9)

5. Since $k=2$ so we found 2 centers

Point	$d(A)$	$d(D)$	Assigned Cluster
B(2,2)	2.83	9.9	A
C(8,8)	11.31	1.41	D
E(1,8)	8.06	8.06	A or D

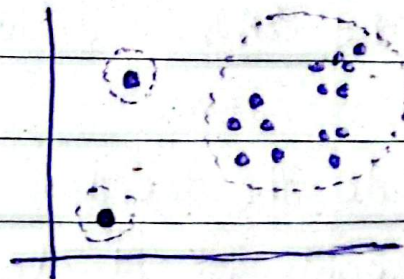
↖
 Smallest distance
 to assign krdenge

• Drawbacks

↳ Sensitive to outliers

↳ Each point can only belong to one cluster

↳ Diff starting point can lead to diff clusters



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▷ K-MEANS CLUSTERING

$O(n \cdot k \cdot \text{iterations})$

↳ The Lloyd Algorithm (Hard Clustering Algo)

Points: $A(0,0)$, $B(2,2)$, $C(8,8)$, $D(9,9)$

$K=2$

1. First pick "K" random centers

$A(0,0)$, $B(2,2)$

2. Assign each point to its nearest center

Point	$d(A)$	$d(B)$	Cluster
$A(0,0)$	0	2.83	A
$B(2,2)$	2.83	0	B
$C(8,8)$	11.31	8.49	B
$D(9,9)$	12.73	9.90	B

Cluster 1: $\{A\}$, Cluster 2: $\{B, C, D\}$

3. Update the centers by taking mean of all the points in cluster.

Cluster 1: $(0/1, 0/1) = (0,0)$

Cluster 2: $(2+8+9/3, 2+8+9/3) = (6.33, 6.33)$

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4. New Centers : $(0,0)$, $(6.33, 6.33)$

5. Reassign the points

Point	$d(0,0)$	$d(6.33, 6.33)$	Cluster
A(0,0)	0	8.95	A
B(2,2)	2.83	6.13	A
C(8,0)	11.31	2.36	B
D(9,9)	12.73	3.77	B

New Clusters:

Cluster 1: {A, B} , Cluster 2: {C, D}

Center 1: $(0+2/2, 0+2/2)$ $(8+9/2, 8+9/2)$

$(1, 1)$

$(8.5, 8.5)$

Reassign Reassign again. If no change in cluster assignment it means DONE!

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▷ SOFT CLUSTERING ALGORITHM

↳ Soft K-Means clustering

→ In hard clustering algo each point belongs to only ONE cluster but

in Soft clustering a point can belong to multiple clusters with probabilities

⇒ ~~Make~~

Points: $x_1(1,1)$, $x_2(2,1)$, $x_3(5,4)$, $x_4(6,5)$

$K = 2$

Initial Centers: $c_1(1,1)$, $c_2(5,4)$ [Given or choose randomly]

$\beta = 1$ (stiffness parameter) (higher $\rightarrow$ harder clustering)

1. Compute distances of point to center

(Euclidean dist)

Point	$d(c_1)$	$d(c_2)$
$x_1(1,1)$	0	5
$x_2(2,1)$	1	4.24
$x_3(5,4)$	5	0
$x_4(6,5)$	6.4	1.41

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2. Now compute memberships using formula

$$w_{i,j} = \frac{e^{-\beta \cdot d(x_i, c_j)}}{\sum_k e^{-\beta \cdot d(x_i, c_k)}}$$

→ this is the same thing we calculated in previous table
 ↪ simple normalization step

Point	$e^{-d(x_i, c_1)}$	$e^{-d(x_i, c_2)}$
$x_1 (1,1)$	$e^0 = 1$	$e^{-5} = 0.0067$
$x_2 (2,1)$	$e^{-1} = 0.367$	$e^{-4.24} = 0.014$
$x_3 (5,4)$	$e^{-5} = 0.007$	$e^0 = 1$
$x_4 (6,5)$	$e^{-6.4} = 0.0017$	$e^{-1.41} = 0.244$

Point	$w_{i,1}$	$w_{i,2}$
x_1	$\frac{1}{1+0.0067} = 0.993$	$\frac{0.0067}{1+0.0067} = 0.007$
x_2	$\frac{0.367}{0.367+0.014} = 0.96$	$\frac{0.014}{0.367+0.014} = 0.04$
x_3	0.007	0.993
x_4	0.007	0.993

Membership Matrix

	C_1	C_2
x_1	0.993	0.007
x_2	0.960	0.040
x_3	0.007	0.993
x_4	0.007	0.993

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3. Update Centers

→ has ek data point ko uski w_{ij} ki value se (x) karna hai

$$c_j = \frac{\sum w_{ij} \cdot x_i}{\sum w_{ij}}$$

For c_1 :

$$\begin{aligned} x \text{ coordinate} &= \frac{(0.993 \times 1) + (0.96 \times 2) + (0.007 \times 5) + (0.07 \times 6)}{0.993 + 0.96 + 0.007 + 0.07} \\ &= 2.99 / 1.967 \\ &= 1.52 \end{aligned}$$

$$\begin{aligned} y \text{ coordinate} &= \frac{(0.993 \times 1) + (0.96 \times 1) + (0.007 \times 4) + (0.07 \times 5)}{0.993 + 0.96 + 0.007 + 0.07} \\ &= 2.016 / 1.967 \\ &= 1.02 \end{aligned}$$

New c_1 (1.52, 1.02)For c_2 :

$$x \text{ coordinate} : 11.01 / 2.033 = 5.41$$

$$y \text{ coordinate} : 8.984 / 2.033 = 4.42$$

New c_2 (5.41, 4.42)

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▷ HIERARCHICAL CLUSTERING

D:

	A	B	C	D
A	0	1	5	6.4
B		0	4.24	5.66
C			0	1.41
D				0

1. Find minimum distance

$$d(A, B) = 1$$

2. Merge A & B.

	A, B	C	D
A, B	0	4.62	6.03
C		0	1.41
D			0

$\frac{5 + 4.24}{2}$
 $\frac{6.4 + 5.66}{2}$

Minimum distance (C, D)

	A, B	C, D
A, B	0	5.325
C, D	5.325	0

$\frac{4.62 + 6.03}{2}$
 5.325

Final Cluster {A, B, C, D} → merge all

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→ When merging two clusters either we can take the avg plus two clusters which we have done in prev ex.

Or we can take minimum plus two clusters.